

HERSCHEL ISLAND, YT, Canada

WMO: 715010

Lat: 69.568N

Lon: 138.911W

Elev: 4

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-32.5	-29.6	-39.4	0.6	-32.5	-36.0	0.7	-29.7	51.6	-0.6	45.4	1.5	7.2	320	1.034

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	62.7	57.1	58.9	54.8	55.8	52.7	58.1	62.0	55.2	58.4	52.9	55.6	8.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.2	67.2	59.8	53.2	60.3	57.0	51.0	55.4	55.0	25.2	62.1	23.3	58.5	21.9	55.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.7	29.9	26.2	DB	-35.4	70.3	3.8	5.3	-38.1	74.1	-40.4	77.3	-42.5	80.2	-45.3	84.1	(4)
(5)			WB	-36.3	60.4	3.7	2.7	-38.9	62.3	-41.0	63.9	-43.1	65.4	-45.8	67.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	-9.7	-10.5	-6.6	8.2	26.2	39.3	48.5	44.3	35.8	22.2	5.0	-5.7		
	DBStd	23.40	12.59	12.74	11.87	11.27	8.10	6.04	6.74	6.68	5.32	8.50	10.72	10.89		
	HDD50	12297	1850	1695	1756	1255	739	326	109	204	425	861	1349	1728		
	HDD65	17678	2315	2115	2221	1705	1204	770	512	643	875	1326	1799	2193		
	CDD50	96	0	0	0	0	0	6	64	26	0	0	0	0		
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0		
	CDH74	9	0	0	0	0	0	0	9	0	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	11.2	12.7	11.3	11.1	9.7	9.9	10.0	10.8	10.4	11.4	12.8	12.3	11.5		
	PrecAvg	10.7	0.4	0.3	0.4	0.3	0.5	1.3	2.0	2.2	1.3	0.9	0.6	0.5		
	PrecMax	13.1	1.2	0.7	1.1	0.9	1.1	2.5	4.3	4.1	2.4	1.6	1.4	1.1		
(15) Precipitation	PrecMin	8.0	0.0	0.0	0.0	0.0	0.1	0.4	0.8	0.8	0.7	0.4	0.1	0.1		
	PrecStd	1.2	0.3	0.2	0.2	0.2	0.2	0.6	0.7	0.7	0.4	0.3	0.3	0.2		
	DB	29.1	26.2	30.5	37.1	45.1	61.4	74.0	65.7	52.6	40.9	33.5	25.6			
	MCWB	27.3	25.9	28.7	34.9	41.2	54.5	65.0	57.7	47.4	38.2	31.7	24.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	22.0	19.2	21.9	32.4	40.6	55.9	65.6	60.3	48.0	36.4	27.9	18.1			
	2% MCWB	20.8	18.3	21.1	31.6	37.7	51.7	60.1	55.2	45.1	34.9	26.6	17.1			
	5% DB	14.4	14.0	16.1	28.6	37.9	51.8	61.0	56.7	45.0	34.0	24.1	13.8			
	5% MCWB	13.7	13.4	15.4	27.7	36.0	48.4	56.4	53.3	43.3	32.8	23.2	13.2			
	10% DB	8.0	8.2	11.0	24.8	35.6	48.2	57.8	53.4	42.8	31.6	20.3	9.3			
	10% MCWB	7.5	7.6	10.4	24.1	34.1	45.5	54.3	51.2	41.4	30.6	19.4	8.7			
	0.4% WB	28.4	25.8	28.5	35.1	42.2	56.2	68.7	59.1	48.2	38.6	30.6	24.7			
	0.4% MCDB	29.3	26.4	30.1	37.0	44.1	59.4	71.4	63.8	51.8	40.6	32.2	25.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	21.3	18.5	21.6	31.9	37.9	51.9	60.6	55.9	45.5	35.2	26.2	16.5			
	2% MCDB	22.6	19.2	22.4	32.7	40.2	56.0	64.4	59.3	47.5	36.3	27.3	17.1			
	5% WB	14.3	13.4	15.7	27.4	36.0	48.5	57.2	53.2	43.6	32.9	22.8	12.5			
	5% MCDB	15.0	14.0	16.4	28.1	37.8	51.5	60.8	56.1	45.0	33.8	23.6	13.1			
	10% WB	8.1	7.7	10.8	23.7	34.2	45.7	54.6	51.1	41.4	30.6	18.9	8.0			
	10% MCDB	8.5	8.2	11.4	24.3	35.4	48.2	57.6	53.4	42.6	31.4	19.8	8.6			
	MDBR	10.2	10.6	10.6	11.2	7.6	9.6	10.5	8.7	6.3	6.6	9.3	9.4			
	5% DB	17.7	18.5	15.5	13.7	12.1	16.5	16.2	14.9	11.1	7.9	11.6	14.4			
(36) Mean Daily Temperature Range	5% WB	17.1	18.8	14.6	12.7	9.5	12.2	10.9	9.8	8.5	6.3	10.9	14.9			
	MCDBR	17.7	18.5	15.1	13.2	11.5	16.1	16.0	14.1	10.8	7.4	11.4	14.8			
	MCWBR	17.2	18.3	14.6	12.2	9.1	12.2	11.4	9.6	8.5	6.3	10.8	14.5			
	MCWBR	17.2	18.3	14.6	12.2	9.1	12.2	11.4	9.6	8.5	6.3	10.8	14.5			
(40) Clear-Sky Solar Irradiance	taub	0.167	0.207	0.242	0.303	0.339	0.349	0.370	0.340	0.301	0.226	N/A	N/A			
	taud	1.886	2.061	2.193	2.082	2.100	2.311	2.386	2.499	2.514	2.197	N/A	N/A			
	Ebn at Noon	10	177	251	261	264	267	256	253	233	167	0	0			
	Edh at Noon	1	13	23	36	40	34	30	24	18	11	0	0			
(44) All-Sky Solar Radiation	RadAvg	4	125	638	1363	1879	1951	1545	987	523	154	0	0			
	RadStd	0	12	33	75	141	154	128	85	50	24	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+1.35	+1.69	N/A	N/A	N/A	N/A	-457	-493	N/A	N/A

Nomenclature: See separate page